Lecture 17

Explicit meromorphic expansions and products

Objectives. *The poles of the cotangent and the zeros of the sine determine explicit series and product formulas. We prove their convergence and normalisation. A finite Beta integral then gives the reciprocal-Gamma product without using a general factorisation theorem.*

By the end of this lecture you will be able to:

- Justify local uniform convergence of the products used here.
- Derive the cotangent expansion by a contour estimate.
- Recover the sine product from its logarithmic derivative.
- Derive the reciprocal-Gamma product from finite Beta integrals.

17.1 Convergence of holomorphic products

An infinite product $\prod_{n\geq 1}(1+u_n(z))$ means the limit of its finite products, in the stated order. A limit of nonzero numbers can be zero, so convergence alone does not describe its zeros. The following criterion also controls the tail away from zero.

Lemma 17.1. *Let u_n be holomorphic on an open set U. Suppose that for every compact $K \subset U$,*

$$\sum_{n=1}^{\infty} \sup_{z\in K} |u_n(z)| < \infty.$$

Then the product converges uniformly on compact sets to a holomorphic function. Locally its zeros, with multiplicity, are exactly the zeros of its factors, provided no factor vanishes identically on the component.

Proof. For complex numbers a_j, expansion of the finite product gives

$$\left|\prod_{j=m}^{n}(1+a_j)-1\right| \leq \prod_{j=m}^{n}(1+|a_j|)-1 \leq \exp\left(\sum_{j=m}^{n}|a_j|\right)-1.$$

The same bound controls all initial products by the exponential of the full sum. Thus their differences tend uniformly to zero on each compact set. The limit is holomorphic by the locally uniform limit result in § 10.6.

On a compact disk inside U, all sufficiently late terms satisfy $|u_n| \leq 1/2$. The power series for the principal logarithm gives $|\log(1+u_n)| \leq 2|u_n|$. Its sum converges uniformly on that disk. The tail product is the exponential of this holomorphic sum and is nowhere zero. The remaining finite product therefore accounts for every zero and its order on a smaller neighbourhood. ∎

For example, $u_n(z) = -z^2/n^2$ satisfies the criterion on every compact subset of $\mathbb{C}$. Its product has simple zeros at every nonzero integer, since precisely one factor vanishes there. It equals one at zero. For $z = 1/2$, the first two factors give the finite value $(1-1/4)(1-1/16) = 45/64$. This finite calculation is not the value of the whole product.

17.2 Partial fractions for the cotangent

The function $\pi\cot(\pi z)$ has a simple pole of residue one at each integer. A sum of individual fractions $1/(z-n)$ would require a summation convention. Pairing positive and negative integers gives convergent terms.

The following contour avoids every integer pole.

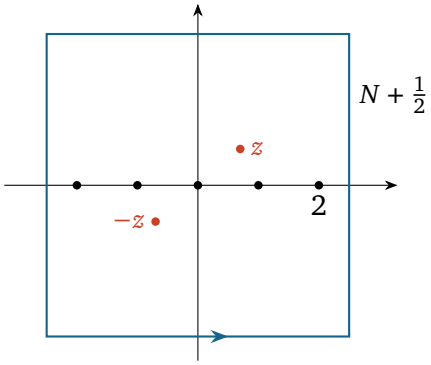


Figure 17.1: A schematic half-integer square avoids the integer cotangent poles. The possible additional poles z and $-z$ lie inside once N is large.

Theorem 17.2. *For $z \notin \mathbb{Z}$,*

$$\pi\cot(\pi z) = \frac{1}{z} + \sum_{n=1}^{\infty} \frac{2z}{z^2 - n^2}. \tag{17.1}$$

The series converges uniformly on compact subsets of $\mathbb{C} \setminus \mathbb{Z}$.

Proof. Fix $z \notin \mathbb{Z}$ and integrate, in the variable ζ, the function

$$h_z(\zeta) = \frac{\pi\cot(\pi\zeta)}{\zeta^2 - z^2}$$

around the positively oriented square with sides $\operatorname{Re}\zeta = \pm(N + 1/2)$ and $\operatorname{Im}\zeta = \pm(N + 1/2)$. Take N large enough to enclose z and $-z$. On the vertical sides $|\cot(\pi\zeta)| = |\tanh(\pi \operatorname{Im}\zeta)| \leq 1$. On the horizontal sides the exponential formula for the cotangent bounds its modulus by $\coth(\pi/2)$, independently of N. On the whole boundary, $|\zeta^2 - z^2| \geq |\zeta|^2 - |z|^2 \geq (N + 1/2)^2 - |z|^2$. The boundary length is $8(N + 1/2)$. The contour integral is therefore $O(1/N)$ and tends to zero.

At an integer n, the residue is $1/(n^2 - z^2)$. The residues at z and $-z$, taken as zero at removable singularities, add to $\pi\cot(\pi z)/z$. The residue theorem, Theorem 11.3, consequently gives in the limit

$$0 = \frac{\pi\cot(\pi z)}{z} - \frac{1}{z^2} + \sum_{n=1}^{\infty} \frac{2}{n^2 - z^2}.$$

Multiplication by z yields the formula. If $|z| \le R$ and $n > 2R$, the magnitude of the nth paired term is at most C_R/n^2. This proves the claimed local uniform convergence away from the finitely many remaining poles. ∎

For instance, at the integer pole $\zeta = 2$, the residue of $h_{1/2}$ is $1/(4 - 1/4) = 4/15$. The apparent singularities at $\zeta = 1/2$ and $\zeta = -1/2$ are removable. Their residues are zero, so only the integer poles contribute in this case.

17.3 The sine product

Theorem 17.3. *For every $z \in \mathbb{C}$,*

$$\sin(\pi z) = \pi z \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2}\right). \tag{17.2}$$

The product converges uniformly on compact sets.

Proof. Let $S(z)$ denote the right-hand side. The product lemma makes S entire with simple zeros exactly at the integers. Away from them, logarithmic differentiation of the finite products and passage to the locally uniform limit gives

$$\frac{S'(z)}{S(z)} = \frac{1}{z} + \sum_{n=1}^{\infty} \frac{2z}{z^2 - n^2} = \pi \cot(\pi z).$$

Derivatives pass to the limit by § 10.6, and division is valid on compact sets avoiding the zeros, where S stays away from zero. The sine has the same logarithmic derivative and the same simple zeros. Thus $\sin(\pi z)/S(z)$ extends holomorphically through every integer, and its derivative is zero elsewhere. The complement of the integers is connected, so the quotient is constant there and hence everywhere. At zero both numerator and denominator are $\pi z + O(z^3)$. The constant is one, proving the product and its normalisation. ∎

The paired factors reflect the paired zeros n and $-n$. Their difference from one is $O(n^{-2})$ on a fixed compact set, which is why no extra exponential factors are needed for this product.

17.4 The reciprocal-Gamma product

Recall from Theorem 13.2 that Γ has no zeros and satisfies

$$B(s,t) = \int_0^1 x^{s-1}(1-x)^{t-1}\,\mathrm{d}x = \frac{\Gamma(s)\Gamma(t)}{\Gamma(s+t)} \quad (\operatorname{Re} s, \operatorname{Re} t > 0).$$

Since Γ has no zeros and only simple poles, $1/\Gamma$ is entire. Define $H_N = \sum_{n=1}^{N} 1/n$. The sequence $H_N - \ln N$ decreases, since $\ln(1 + 1/N) > 1/(N+1)$ by integration of $1/x$ on $[N, N+1]$. Integral comparison also gives $H_N \ge \ln(N+1)$, so the sequence is bounded below. Define the *Euler constant* by

$$\gamma_E = \lim_{N\to\infty} (H_N - \ln N).$$

Its subscript distinguishes it from a contour or a decay rate.

Theorem 17.4. *The reciprocal Gamma function has the locally uniformly convergent product*

$$\frac{1}{\Gamma(s)} = s e^{\gamma_E s} \prod_{n=1}^{\infty} \left(1 + \frac{s}{n}\right) e^{-s/n}, \qquad s \in \mathbb{C}. \tag{17.3}$$

Proof. For $\operatorname{Re} s > 0$, the Beta identity and recurrence give the finite equality

$$B(s, N+1) = \frac{N!}{s(s+1)\cdots(s+N)}.$$

Substitute $x = u/N$ in the integral for $N^s B(s, N+1)$ and extend the integrand by zero beyond N. On $0 < u < N$ its modulus is $u^{\operatorname{Re} s - 1}(1 - u/N)^N \le u^{\operatorname{Re} s - 1} e^{-u}$. For each $u > 0$, the integrand tends to $u^{s-1}e^{-u}$. Dominated convergence proves

$$\frac{N^s N!}{s(s+1)\cdots(s+N)} \longrightarrow \Gamma(s).$$

The limit is nonzero, so reciprocation gives

$$\frac{1}{\Gamma(s)} = \lim_{N\to\infty} s e^{s(H_N - \ln N)} \prod_{n=1}^{N} \left(1 + \frac{s}{n}\right) e^{-s/n}.$$

Here the equality follows by cancelling e^{sH_N} against the finite product of $e^{-s/n}$. On $|s| \le R$, Taylor's series for $e^{-s/n}$ shows

$$\left| \left(1 + \frac{s}{n}\right) e^{-s/n} - 1 \right| \le \frac{C_R}{n^2} \qquad (n > 2R).$$

The product lemma therefore gives an entire limit. Since $H_N - \ln N \to \gamma_E$, the desired identity holds on $\operatorname{Re} s > 0$. Both sides are entire. The identity theorem extends it to $\mathbb{C}$. ∎

The linear terms cancel in each factor. Without $e^{-s/n}$, their logarithms would have a divergent harmonic term. The initial factor s gives the zero at zero, and the nth factor gives the zero at $s = -n$. Each is simple, in agreement with the Gamma poles.

17.5 General existence theorems

A subset of $\mathbb{C}$ is *discrete* here if it has no finite accumulation point. An entire function other than zero has a discrete zero set by the identity theorem. The converse is an existence theorem.

Theorem 17.5 (Weierstrass · *cited*)**.** *For any discrete set of points in $\mathbb{C}$ and any assigned positive integer multiplicity at each point, there is an entire function whose zeros are exactly those points with those multiplicities.*

Theorem 17.6 (Mittag–Leffler · *cited*)**.** *At each point a of a discrete subset of $\mathbb{C}$, prescribe a finite principal part $\sum_{j=1}^{m_a} c_{a,j}(z-a)^{-j}$. There is a meromorphic function with exactly those principal parts and no other poles.*

For Weierstrass existence see Stein–Shakarchi, *Complex Analysis*, Chapter 5, Theorem 4.1, p. 145. Mittag–Leffler existence is Freitag–Busam, *Complex Analysis*, Theorem IV.3.1. The proofs of these existence statements are external. They require convergence corrections

to arbitrary zero or pole data. The explicit cotangent, sine and reciprocal-Gamma proofs above supplied their own corrections and normalisations. They do not depend on either cited theorem. In particular, specifying zeros alone does not determine an entire function uniquely: multiplication by $e^{h(z)}$ for entire h preserves all its zeros and their orders.

Summary.

- Locally summable factor errors give locally uniform holomorphic products. Their sufficiently late tails are nonzero.
- Pairing the integer poles gives the proved cotangent expansion. That expansion is the logarithmic derivative of the normalised sine product.
- Finite Beta integrals and the Euler constant give the proved reciprocal-Gamma product with factors $(1 + s/n)e^{-s/n}$.
- General zero and principal-part existence are cited separately.

References

E. M. Stein and R. Shakarchi, *Complex Analysis*, 2003, Chapter 5, pp. 142–145, and Chapter 6, Theorem 1.7, pp. 166–167. Their reciprocal-Gamma proof uses general factorisation; the Beta-limit proof here is independent of it. E. Freitag and R. Busam, *Complex Analysis*, 2005, Theorem IV.3.1, pp. 223–227, for Mittag–Leffler existence.